EXACTNESS OF THE LOWER SHRIEK FUNCTOR ON MODULES UNDER FLATNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the degree description comes from Lemma lemma-restriction-module-to-components-site (in-source stub); external tool from Modules on Sites.

Editorial note. The label lemma-exactness-g-shriek-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Degree description of g_{n!}. Same unit, Lemma lemma-restriction-module-to-components-site. Describes g_{n!}G as the module whose degree m part is the direct sum of f_phi^*G, and the assumptions.
- Flat pullback exactness. Modules on Sites, Lemma sites-modules-lemma-flat-pullback-exact. If f_phi^{-1}O_n -> O_m are flat then f_phi^* is exact.

Source reference. Original reference 0D72: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1459–1459. Prerequisite pointer invoked at proof line(s) 1535; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04JC: sources/stacks-chow/tags.txt, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 1553; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With assumptions as in Lemma 0D72 the functor $g_{n!} : Mod(\mathcal{O}_n) \to Mod(\mathcal{O})$ is exact if the maps $f_\varphi^{-1}\mathcal{O}_n \to \mathcal{O}_m$ are flat for all $\varphi : [n] \to [m]$.*

Proof. Recall that $g_{n!}\mathcal{G}$ is the $\mathcal{O}$-module whose degree m part is the $\mathcal{O}_m$-module

$$\bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^*\mathcal{G}$$

Here the morphism of ringed topoi $f_\varphi : (Sh(\mathcal{C}_m), \mathcal{O}_m) \to (Sh(\mathcal{C}_n), \mathcal{O}_n)$ uses the map $f_\varphi^{-1}\mathcal{O}_n \to \mathcal{O}_m$ of the statement of the lemma. If these maps are flat, then f_φ^* is

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exact (Modules on Sites, Lemma 04JC). By definition of the site $\mathcal{C}_{total}$ we see that these functors have the desired exactness properties and we conclude. $\square$